

# ME315\_Lab03

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```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(leaflet)
```

1.

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA\_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airports <- read_csv(
  "airports.csv", col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

Rows: 322 Columns: 5

-- Column specification -----

Delimiter: ","

chr (3): IATA\_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
partes <- list()
```

```
callback <- DataFrameCallback$new(function(x, pos) {
```

```
  x <- x[, c("DESTINATION_AIRPORT", "ARRIVAL_DELAY")]
```

```
  x <- x[
    !grepl("^1", x$DESTINATION_AIRPORT) &
    !is.na(x$DESTINATION_AIRPORT) &
    !is.na(x$ARRIVAL_DELAY),]

```

```
  estadisticas <- x %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      n = n(),
      soma = sum(ARRIVAL_DELAY),
      .groups = "drop")

```

```
  partes[[length(partes) + 1]] <- estadisticas})
```

```
flights <- read_csv_chunked(
  "flights.csv", callback = callback, chunk_size = 1000000)
```

-- Column specification -----

cols(

.default = col\_double(),

AIRLINE = col\_character(),

TAIL\_NUMBER = col\_character(),

```

    ORIGIN_AIRPORT = col_character(),
    DESTINATION_AIRPORT = col_character(),
    SCHEDULED_DEPARTURE = col_character(),
    DEPARTURE_TIME = col_character(),
    WHEELS_OFF = col_character(),
    WHEELS_ON = col_character(),
    SCHEDULED_ARRIVAL = col_character(),
    ARRIVAL_TIME = col_character(),
    CANCELLATION_REASON = col_character()
)
i Use `spec()` for the full column specifications.

```

```

estatisticas_finais <- bind_rows(partes) %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(n = sum(n), soma = sum(soma), .groups = "drop") %>%
  mutate(atraso_medio = soma / n)

estatisticas_finais

```

```

# A tibble: 322 x 4
  DESTINATION_AIRPORT      n  soma atraso_medio
  <chr>              <int> <dbl>         <dbl>
1 ABE                2220 12874          5.80
2 ABI                2224  9077          4.08
3 ABQ             18953 106415          5.61
4 ABR                 657 -2227         -3.39
5 ABY                 864  7501          8.68
6 ACK                 487  2393          4.91
7 ACT                1541 16392         10.6
8 ACV                1267 12259          9.68
9 ACY                3530 41744         11.8
10 ADK                 89  -284         -3.19
# i 312 more rows

```

2.

Na tabela flights, a chave que identifica o aeroporto de destino é: DESTINATION\_AIRPORT

Na tabela airports, a chave correspondente é: IATA\_CODE

Portanto, a relação é:

`flights`  $\xrightarrow{\text{DESTINATION\_AIRPORT}}$  `airports`  $\xrightarrow{\text{IATA\_CODE}}$

```
aeroportos_ausentes <- flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE")) %>%
  distinct(DESTINATION_AIRPORT)
aeroportos_ausentes
```

```
# A tibble: 0 x 1
# i 1 variable: DESTINATION_AIRPORT <chr>
```

```
flights <- flights %>%
  left_join(airports %>%
    select(IATA_CODE, CITY, STATE), by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

3.

```
aeroportos_por_estado <- airports %>%
  count(STATE, name = "numero_aeroportos") %>%
  arrange(desc(numero_aeroportos))

aeroportos_por_estado
```

```
# A tibble: 54 x 2
  STATE numero_aeroportos
  <chr>          <int>
1 TX              24
2 CA              22
3 AK              19
4 FL              17
5 MI              15
6 NY              14
7 CO              10
8 MN               8
9 MT               8
10 NC              8
# i 44 more rows
```

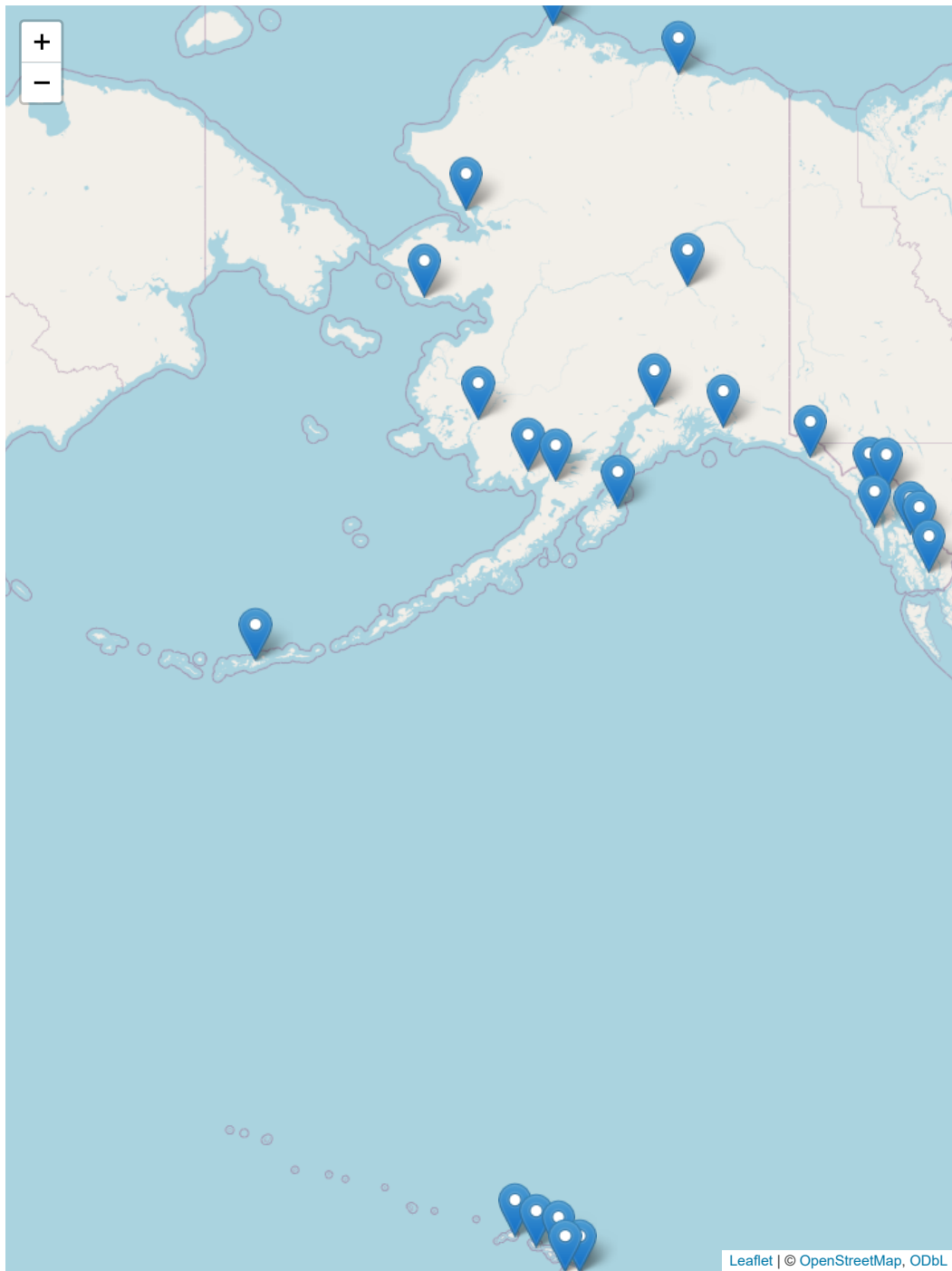
4.

```
mapa_dados <- estatisticas_finais %>%
  left_join(
    airports,
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
```

```
) %>%  
filter(!is.na(LATITUDE), !is.na(LONGITUDE))
```

```
mapa <- leaflet(mapa_dados) %>%  
  addTiles() %>%  
  addMarkers(lng = ~LONGITUDE, lat = ~LATITUDE, popup = ~paste0(  
    "<b>Aeroporto:</b> ", DESTINATION_AIRPORT,  
    "<br><b>Cidade:</b> ", CITY,  
    "<br><b>Estado:</b> ", STATE,  
    "<br><b>Atraso médio:</b> ", round(atraso_medio, 2), " minutos"))  
mapa
```

Warning in file.info(file): cannot resolve owner of file  
'\\smb\ra281972\Downloads\ME315\file1bd86c914db2\webshot1bd85a651b4a.pdf': Não  
foi feito mapeamento entre os nomes de conta e as identificações de segurança



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 18:38:27 -03"
```